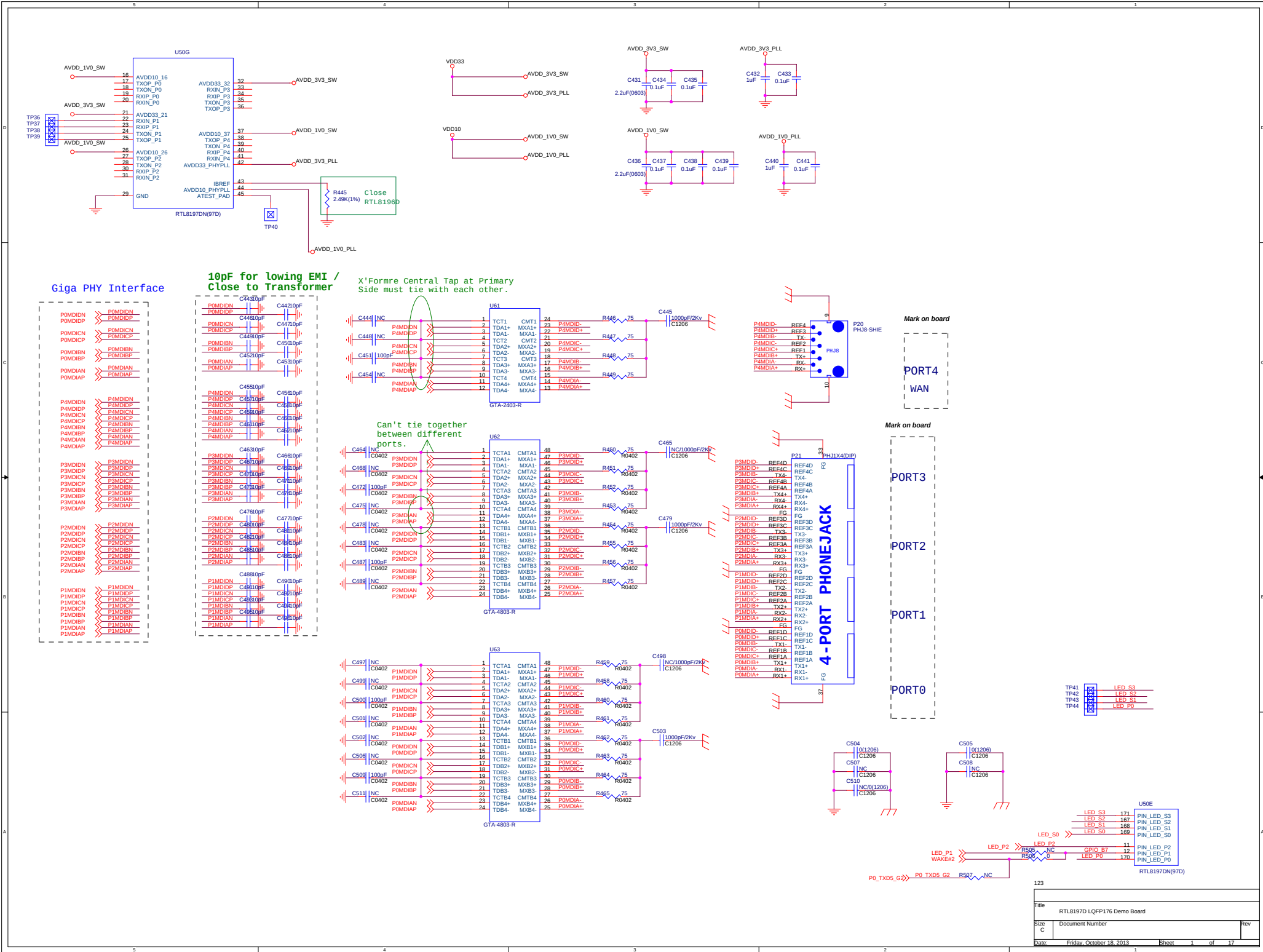
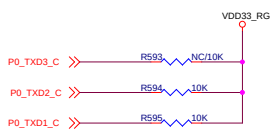
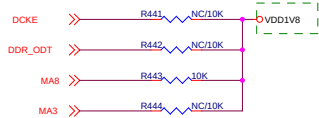
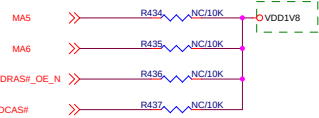
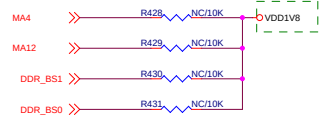
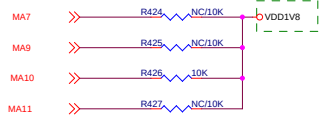






PIN_M_A7	phyidsel[0]/MIIL_SLV	NFB_I_PHY_ID	
		10=01000, 11=10000	
PIN_M_A9	ck_m2x_freq_sel[0]	ck_freq_sel[2:0] (MHZ)	
		000:	156.25
		001:	193.75
		010:	181.25
PIN_M_A10	ck_m2x_freq_sel[1]	100:	Reserve
		101:	125
PIN_M_A11	ck_m2x_freq_sel[2]	110:	Reserve
		111:	168.75

PIN_M_A12	sel_40m	sel_40m	
		1=40MHZ	
		0=25MHZ(default)	
PIN_M_A4	ENABLE_EXT_RSTN	ENABLE_EXT_RSTN	
		1=enable	
		0=disable	
PIN_M_A16	ck_cpu_freq_sel[3]	ck_cpu_freq_sel[3:1] (MHZ)	
		000:	660
		001:	Reserve
		010:	Reserve
PIN_M_A15	ck_cpu_freq_sel[2]	100:	620
		101:	580
		110:	540
		111:	500
PIN_U0_TX	ck_cpu_freq_sel[1]		

PIN_LED_S0	swap bus/one-word	
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PIN_M_A5	dbg_olt_mode[1]	dbg_olt_mode[1:0]	
PIN_M_A6	dbg_olt_mode[0]		
DRAS#_OE_N	mii_tx_dly[0]		
DRAS#_OE_N	mii_rx_dly[2]		
		11: strap_ctl_sys_dbg_sel(default)	

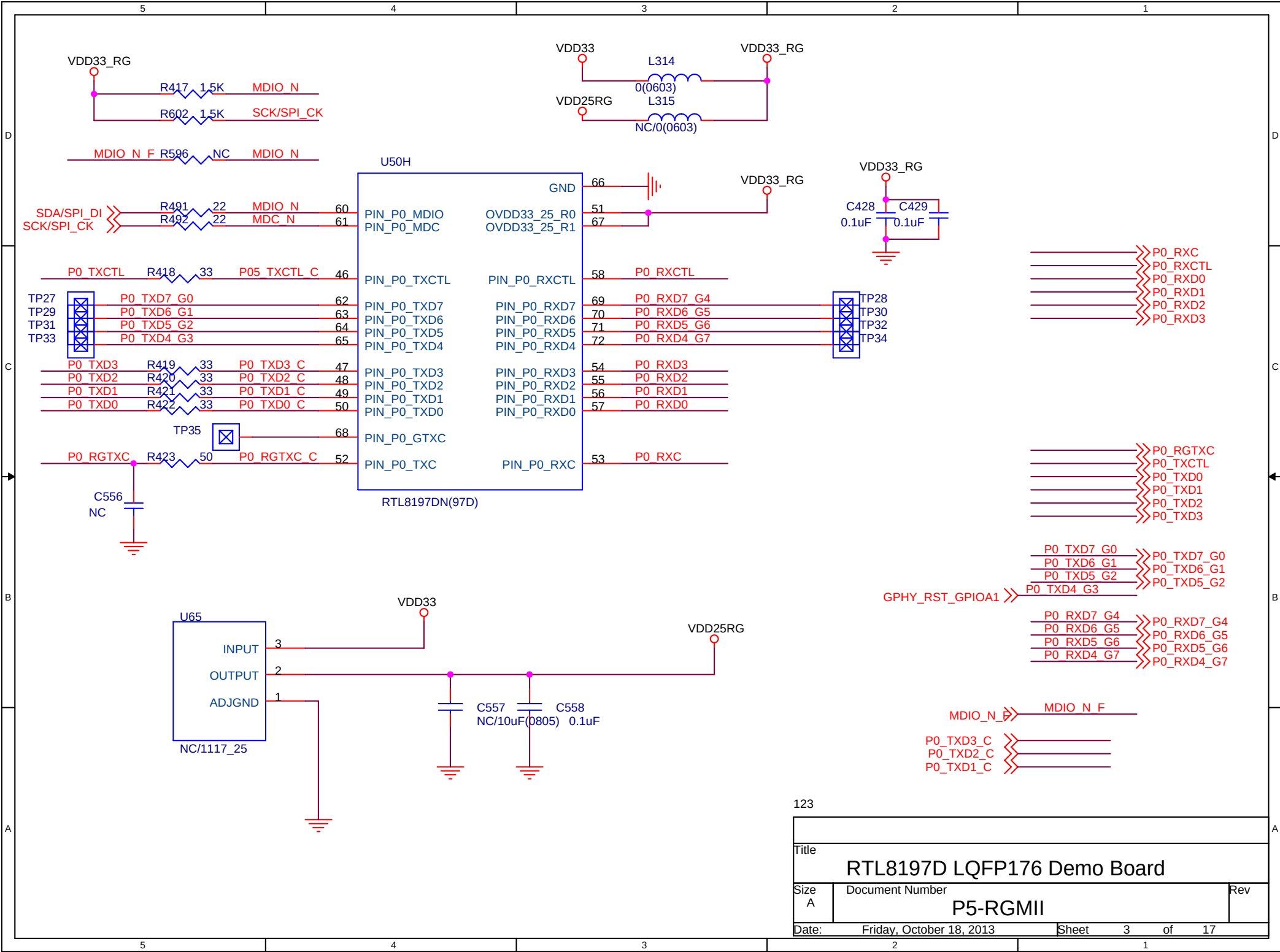
pin name	function name	function detail	
PIN_M_A0	boot_sel[0]	boot_select[2:0]	
PIN_M_A1	boot_sel[1]	000:	SPI
		001:	NOR
PIN_M_A2	boot_sel[2]	100:	ROM1
		101:	ROM2
		110:	ROM3
		111:	ROM4

PIN_M_CKE	strap_test_mode	strap_test_mode	
		1=TEST mode	
		0=normal mode(default)	
MA14_ODT	strap_test_mode	swap bus/byte	

PIN_M_A8	DRAM_TYPE[1]	DRAM_TYPE[1]	
		1=DDR	
		0=SDR (Default)	
PIN_M_A3	DRAM_TYPE[0]_INV	DRAM_TYPE[0]_INV	
		1=DDR1	
		0=DDR2 (Default)	

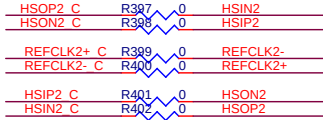
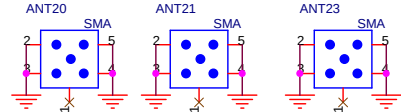
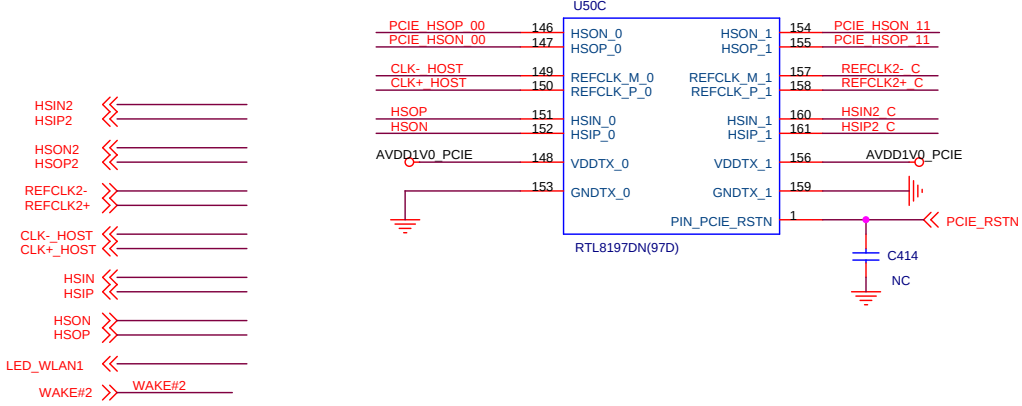
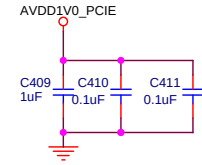
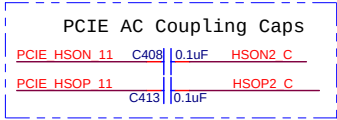
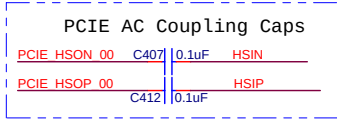
		NAFC_AC[1:0]	
		00: 3 addr cycle	
		01: 4 addr cycle	
		10: 5 addr cycle	
		11: reserved	

PIN_P0_TXD3	NAFC_AC[0]/mii_mode	NAFC_RC	
PIN_P0_TXD2	NAFC_AC[1]/mii_mode		
PIN_P0_TXD1	NAFC_RC/mii_rx_dly[1]		
		0: 1 cycle cmd (512 Byte/page)	
		1: 2 cycle cmd (2048 Byte/page)	

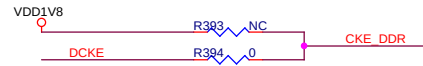
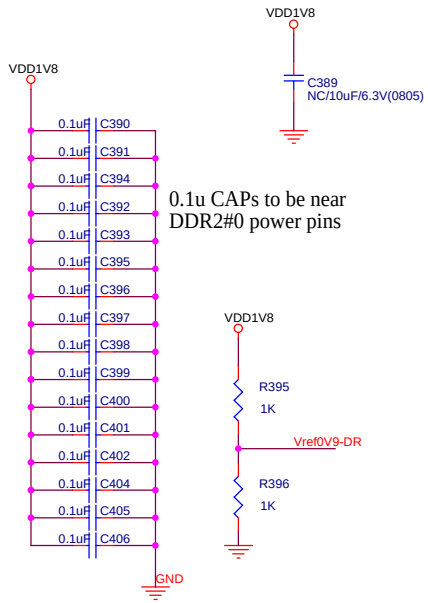
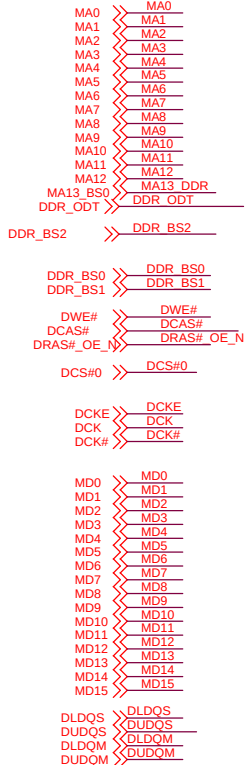


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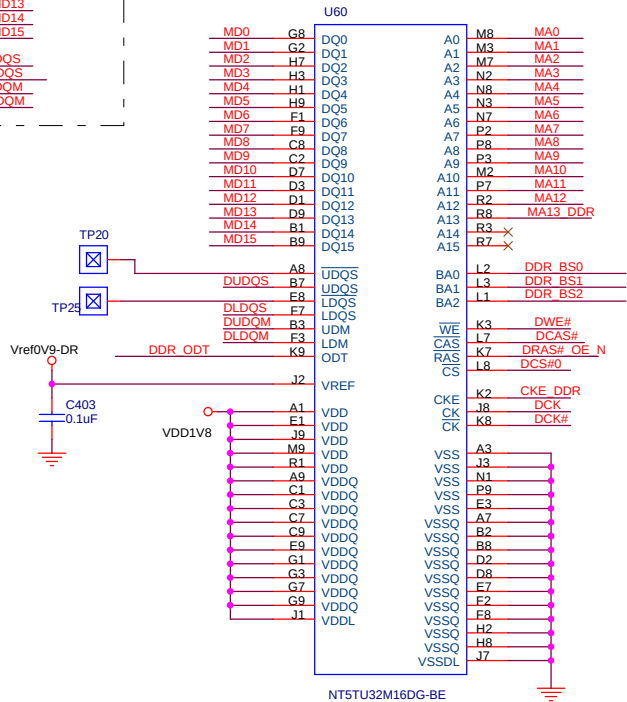
Title		
RTL8197D LQFP176 Demo Board		
Size	Document Number	Rev
A	P5-RGMII	
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RTL8197D ---->
DDR2



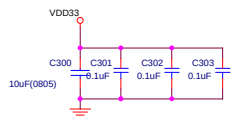
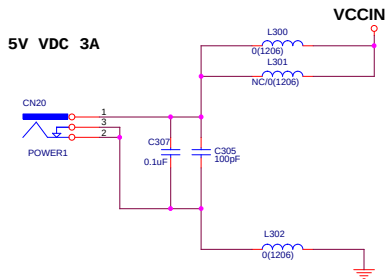
DDR2上在正面



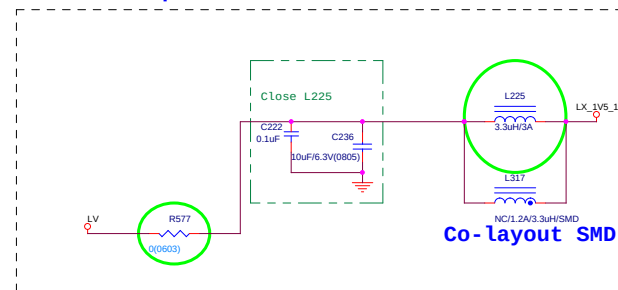
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Title		
RTL8197D LQFP176 Demo Board		
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Custom	DDR2	
Date:	Friday, October 18, 2013	Sheet 5 of 17

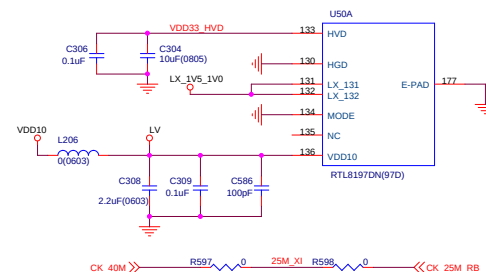
5V VDC 3A



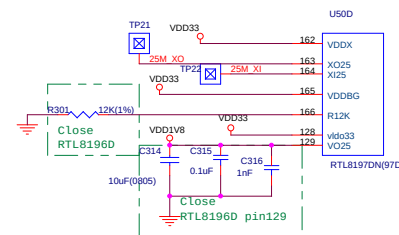
Option: VDD10 Internal SWR



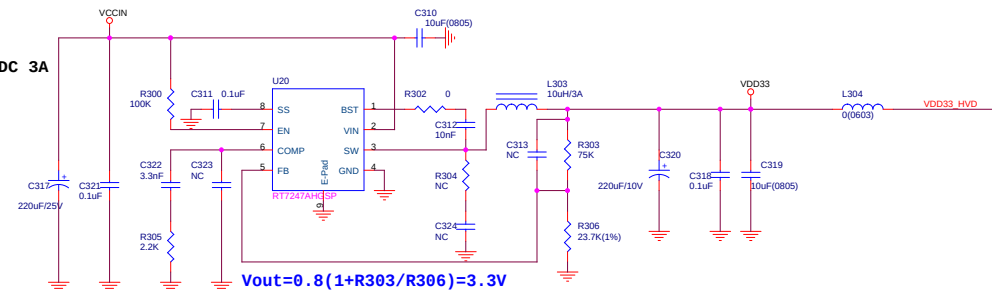
Co-layout SMD



CK_40M >> R597 25M_XI R598 25M_RB <<

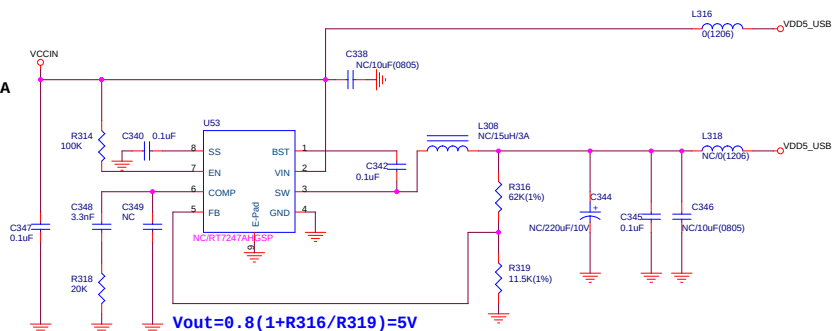


5V VDC 3A



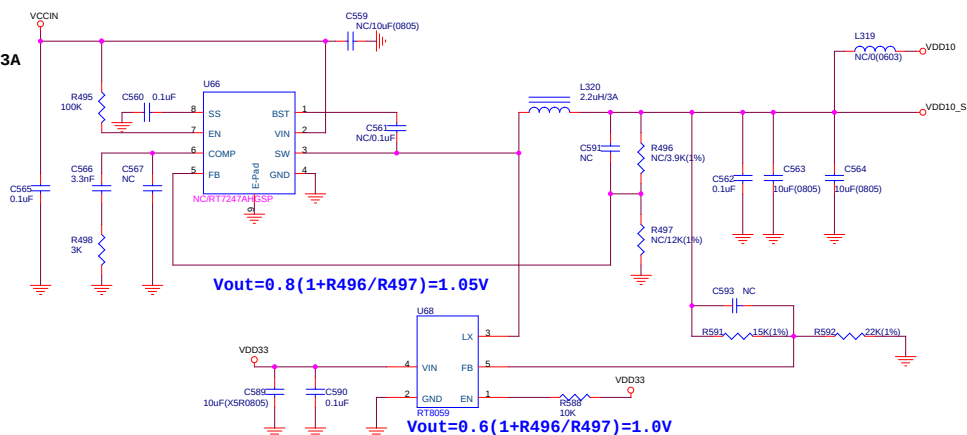
$$V_{out} = 0.8(1 + R303/R306) = 3.3V$$

5V VDC 3A



$$V_{out} = 0.8(1 + R316/R319) = 5V$$

5V VDC 3A

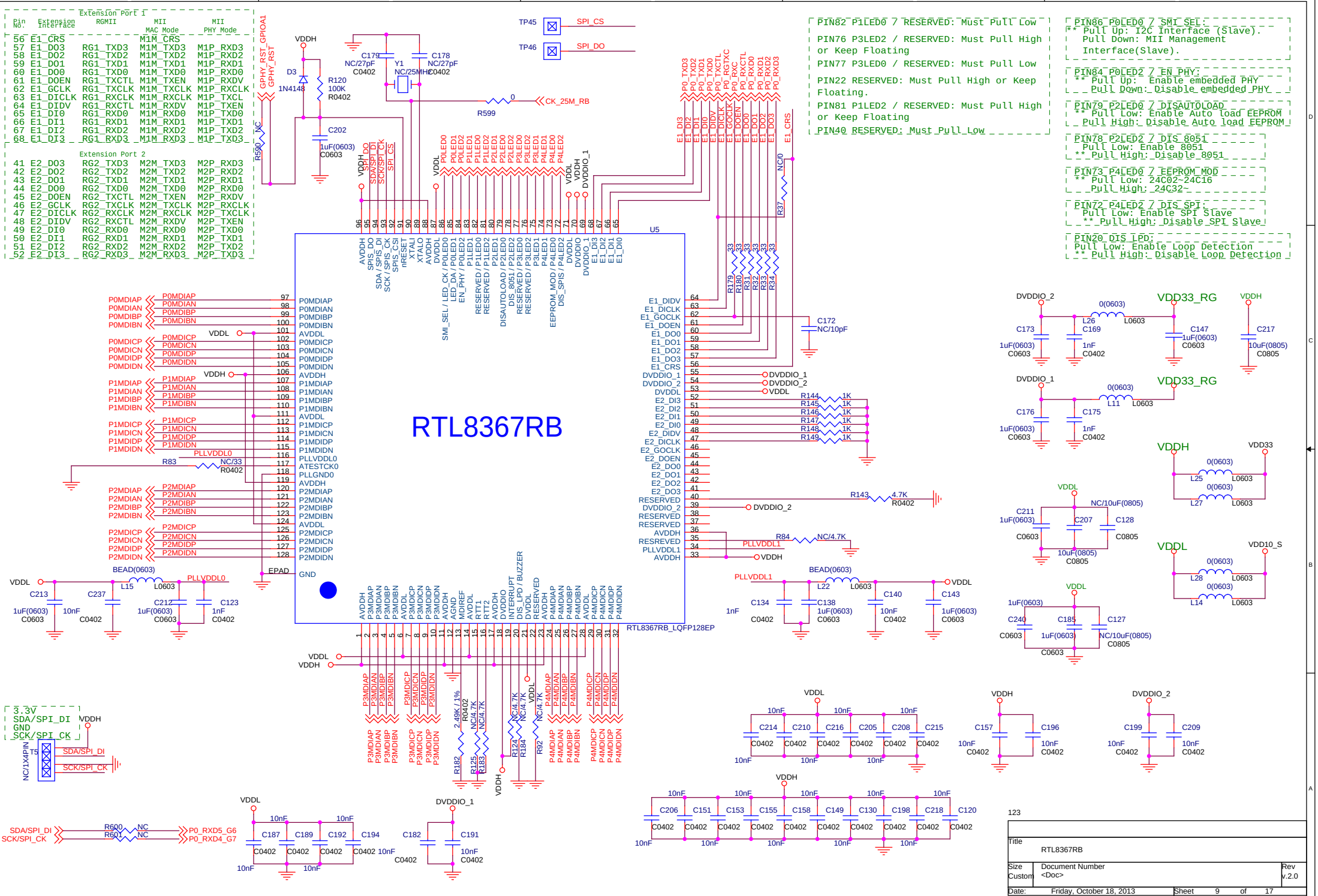


$$V_{out} = 0.8(1 + R496/R497) = 1.05V$$

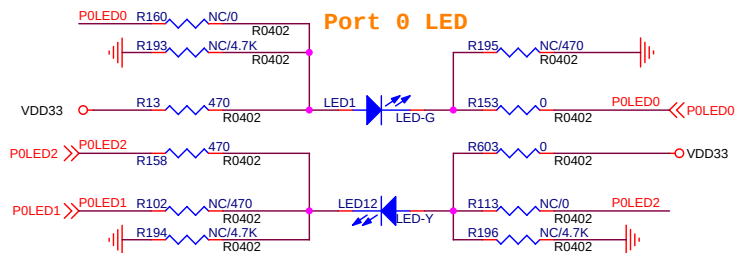
$$V_{out} = 0.6(1 + R496/R497) = 1.0V$$

123

Title				
RTL8197D LQFP176 Demo Board				
Size	Document Number			Rev
C	SYSTEM POWER			
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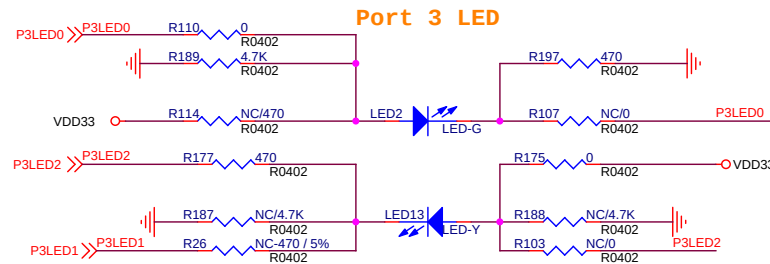


Title			
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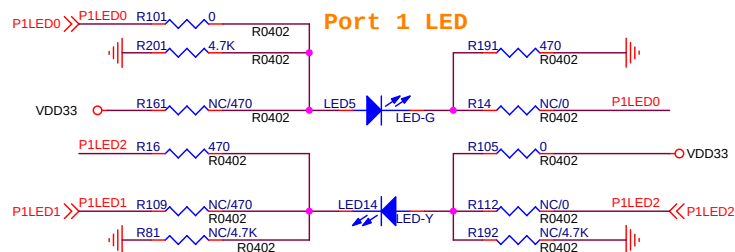
P0LED0 / SMI_SEL
 ** Pull Up: I2C Interface(Slave)
 Pull Down: MII Management
 Interface(Slave)

P0LED2 / EN_PHY:
 ** Pull Up: Enable embedded PHY
 Pull Down: Disable embedded PHY



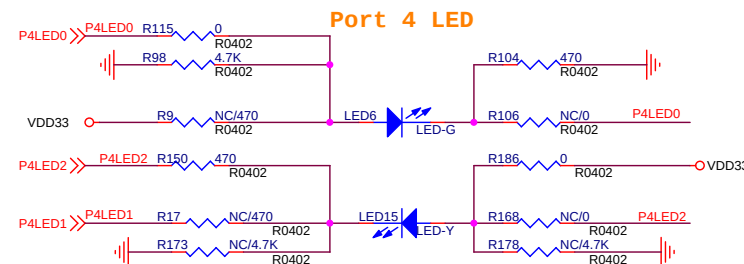
P3LED0 / RESERVED: Must
 Pull Low

P3LED2 / RESERVED: Must Pull
 High or Keep Floating



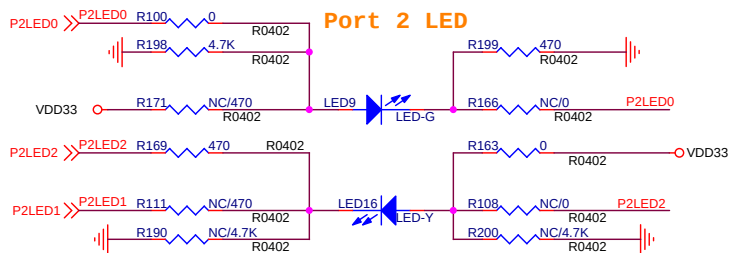
P1LED0 / RESERVED:
 Must Pull Low

P1LED2 / RESERVED: Must Pull High
 or Keep Floating



P4LED0 / EEPROM_MOD
 ** Pull Low: 24C02~24C16
 Pull High: 24C32~24C256

P4LED2 / DIS_SPI:
 Pull Low: Enable SPI Slave
 ** Pull High: Disable SPI Slave

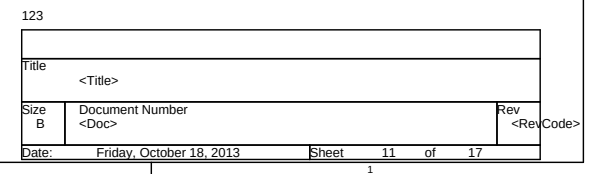


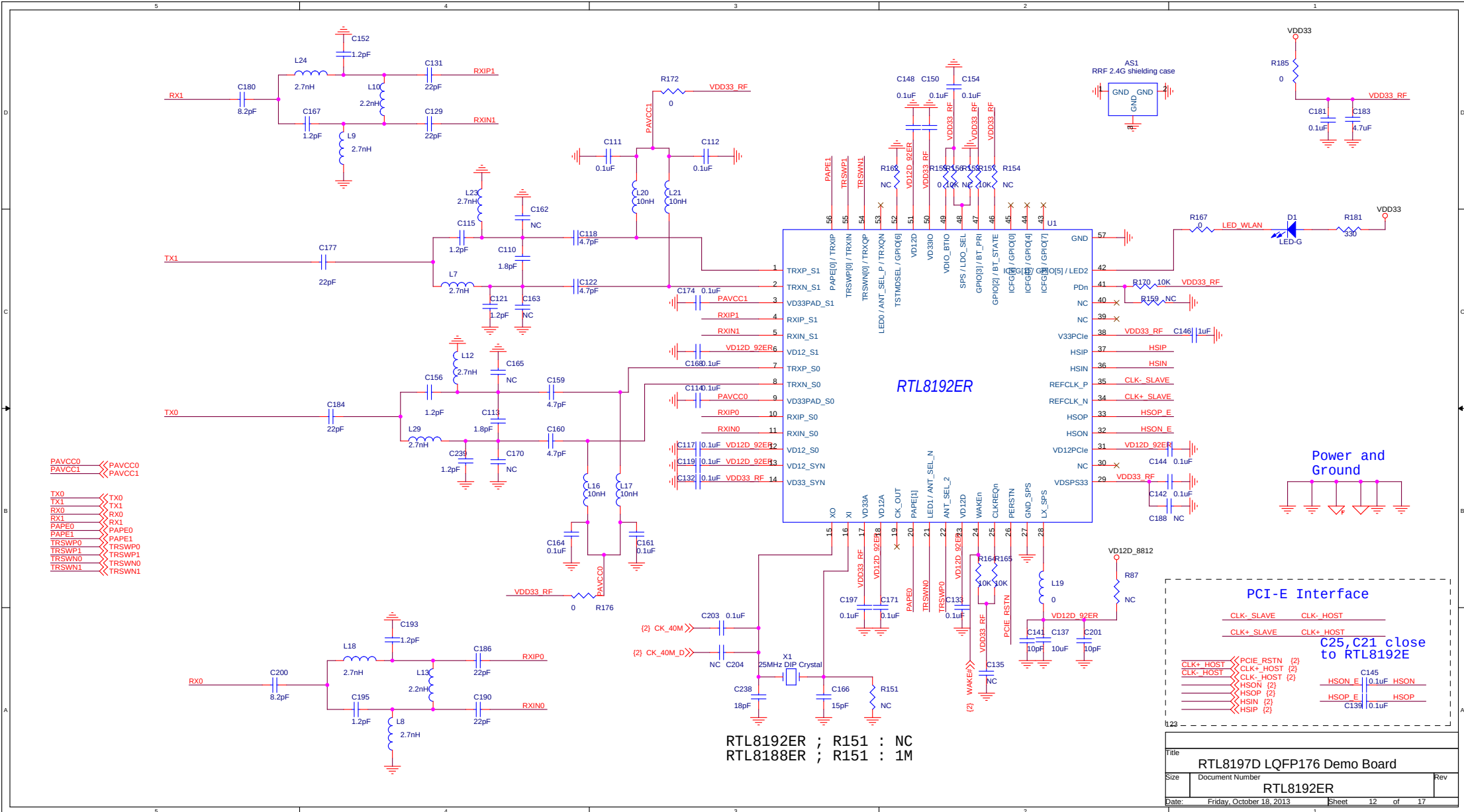
P2LED0 / DISAUTOLoad
 ** Pull Low: Enable Auto
 load EEPROM
 Pull High: Disable Auto
 load EEPROM

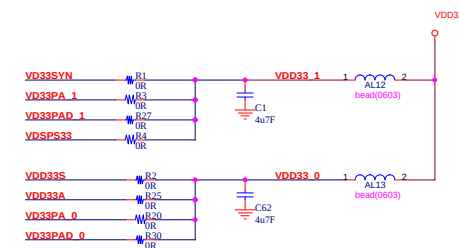
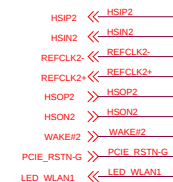
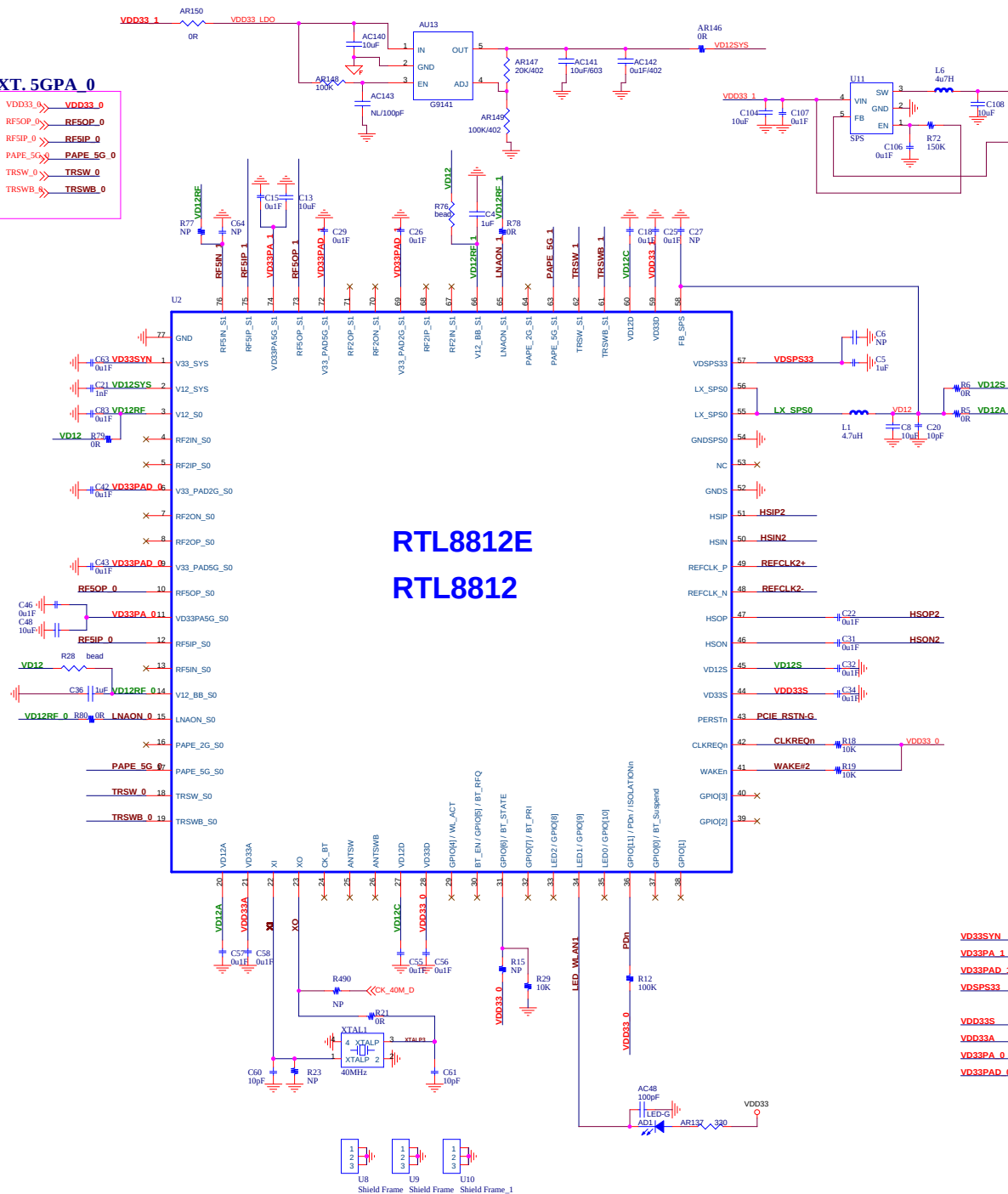
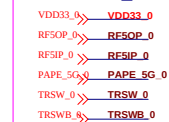
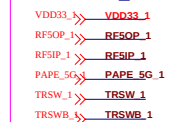
P2LED2 / DIS_8051
 Pull Low: Enable 8051
 ** Pull High: Disable 8051

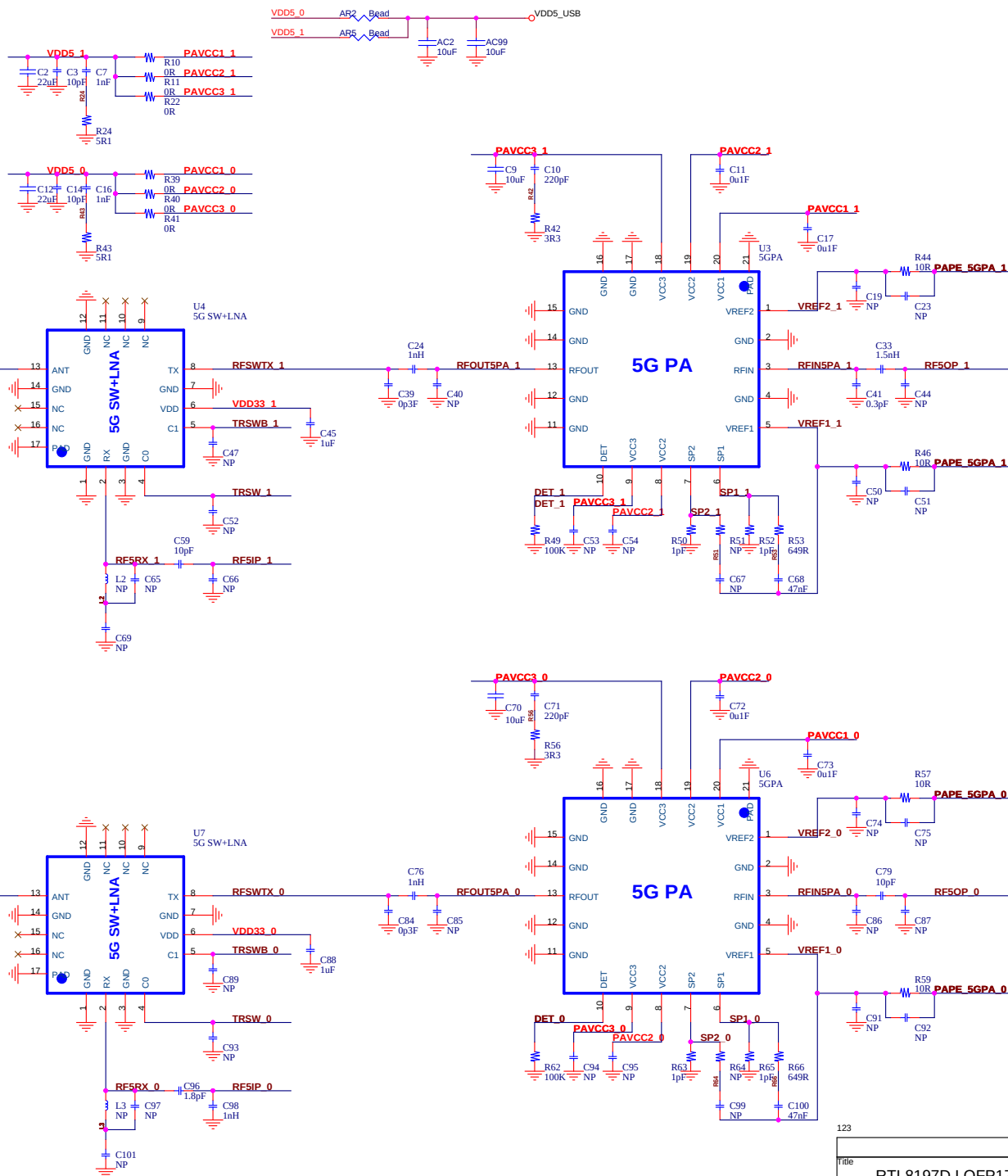
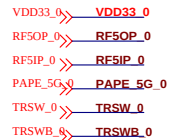
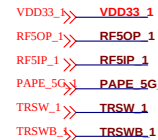
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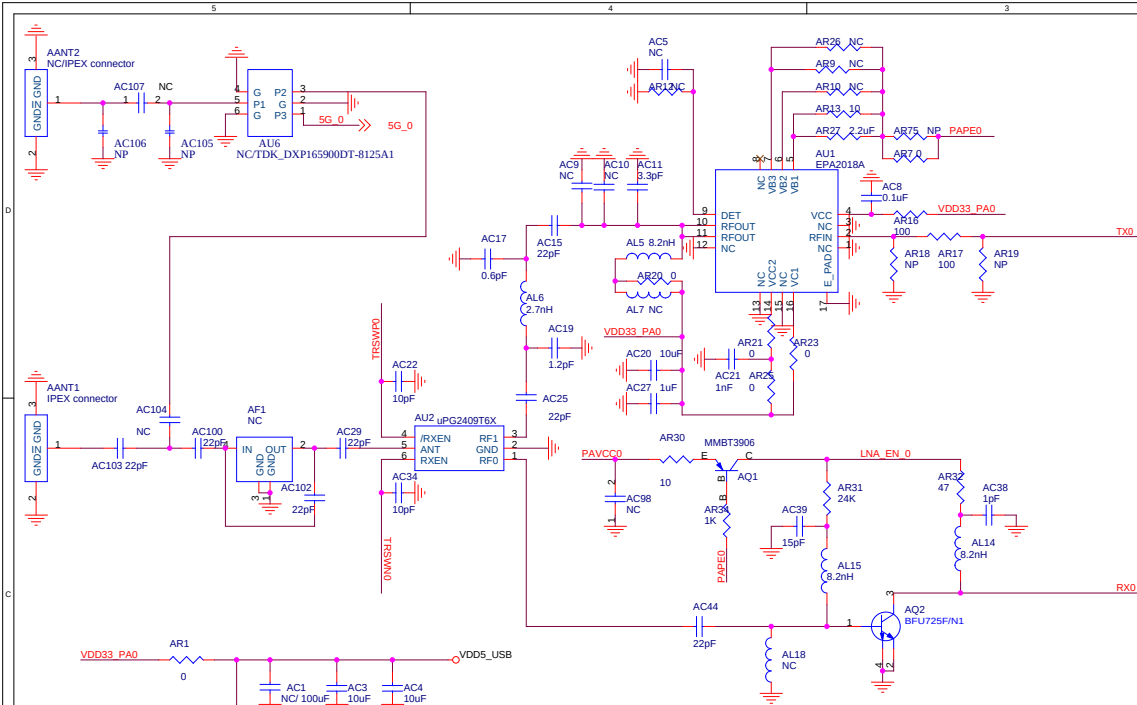
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RTL8367RB LED		
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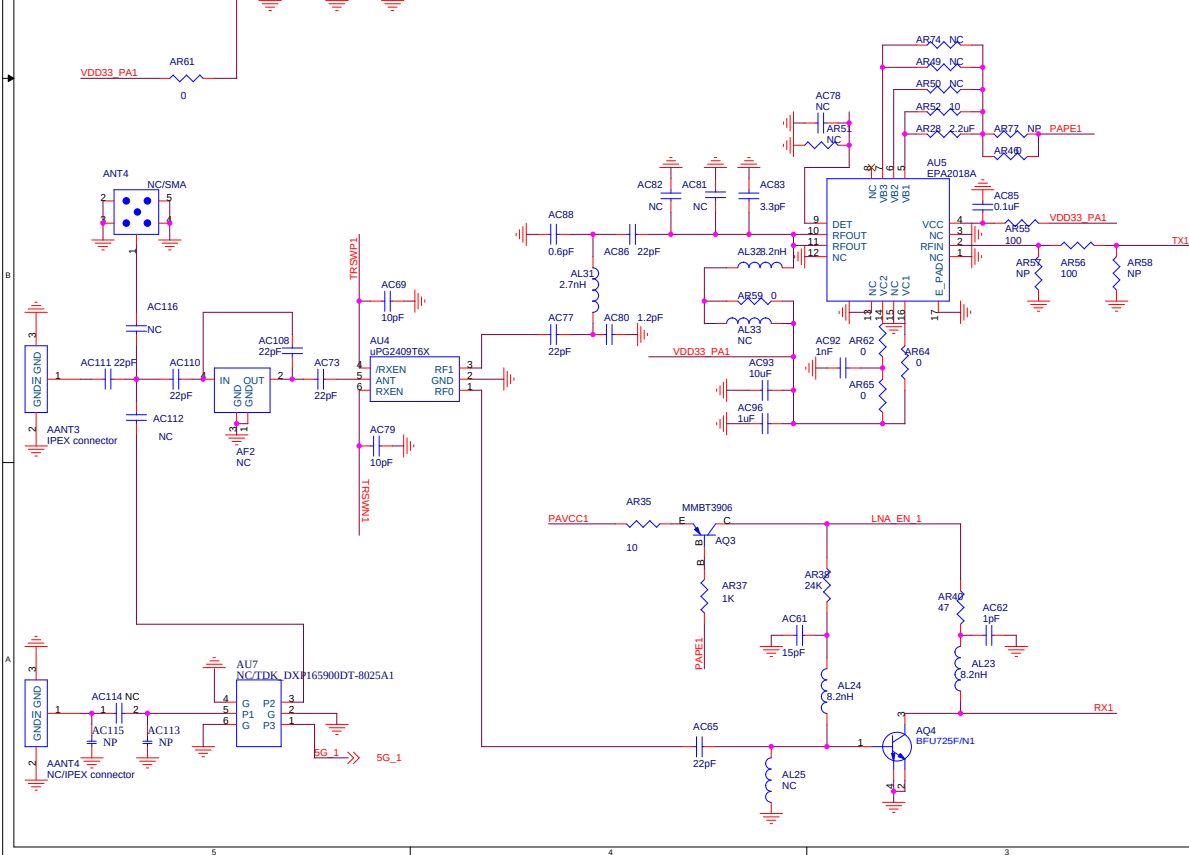




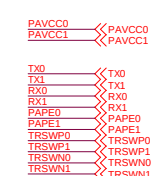




PATH A		
AU1	LX5518	EPA2018A
AR9	412R	NC
AR10	120R	NC
AR12	49.9K	NC
AR13	412R	10R
AR26	1uF	NC
AR27	NC	2.2uF
AR23	20R	0R
AR25	5.1R	0R
AC5	10pF	NC
AC21	12pF	1nF
AR25	5.1R	0R
AL7	8.2nH	NC



PATH B		
AU5	LX5518	EPA2018A
AR49	412R	NC
AR50	120R	NC
AR51	49.9K	NC
AR52	412R	10R
AR74	1uF	NC
AR28	NC	2.2uF
AR64	20R	0R
AR65	5.1R	0R
AC78	10pF	NC
AC92	12pF	1nF
AL33	8.2nH	NC
AR62	5.1R	0R



5	4	3	2	1
D				D
C				C
B				B
A				A

RTL8197DN(97D) GPIO definition

GPIO name	I/O	Function
GPIOA6	O	SYSTEMLED/WPSLED
GPIOA4	I	WAKE#
GPIOA5	I	Reset/Default
GPIOA3	I	WPS
GPIOA1	O	RTL8167RB RESET
GPIOA2	O	PCIE1 RESET

123

Title

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